

Trigonometric Identities & Trigonometric Integrals

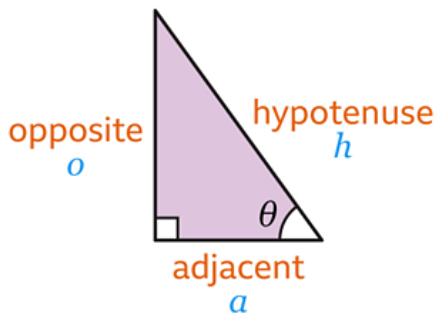
MATH 146 – Calculus II

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Right Triangle Trigonometry



For this definition, we assume

$$0 \leq \theta \leq \frac{\pi}{2} \text{ or } 0^\circ \leq \theta \leq 90^\circ.$$

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$$

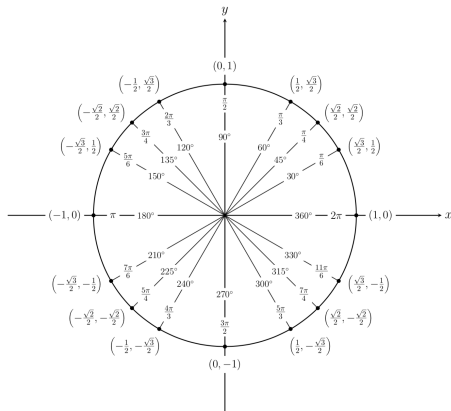
$$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$$

$$\sec(\theta) = \frac{\text{hypotenuse}}{\text{opposite}}$$

$$\csc(\theta) = \frac{\text{hypotenuse}}{\text{adjacent}}$$

$$\cot(\theta) = \frac{\text{adjacent}}{\text{opposite}}$$

Right Triangle Trigonometry



For this definition, θ can take any value.

$$\sin(\theta) = y$$

$$\cos(\theta) = x$$

$$\tan(\theta) = \frac{y}{x}$$

$$\sec(\theta) = \frac{1}{x}$$

$$\csc(\theta) = \frac{1}{y}$$

$$\cot(\theta) = \frac{x}{y}$$

Trigonometric Identities

Tangent and Cotangent Identities

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

$$\cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$$

Reciprocal Identities

$$\sin(\theta) = \frac{1}{\csc(\theta)}$$

$$\csc(\theta) = \frac{1}{\sin(\theta)}$$

$$\cos(\theta) = \frac{1}{\sec(\theta)}$$

$$\sec(\theta) = \frac{1}{\cos(\theta)}$$

$$\tan(\theta) = \frac{1}{\cot(\theta)}$$

$$\cot(\theta) = \frac{1}{\tan(\theta)}$$

Trigonometric Identities

Pythagorean Identities

$$\sin^2(\theta) + \cos^2(\theta) = 1$$

$$\tan^2(\theta) + 1 = \sec^2(\theta)$$

$$1 + \cot^2(\theta) = \csc^2(\theta)$$

Even/Odd Formulas

$$\sin(-\theta) = -\sin(\theta)$$

$$\csc(-\theta) = -\csc(\theta)$$

$$\cos(-\theta) = \cos(\theta)$$

$$\sec(-\theta) = \sec(\theta)$$

$$\tan(-\theta) = -\tan(\theta)$$

$$\cot(-\theta) = -\cot(\theta)$$

Trigonometric Identities

Periodic Formulas

If n is an integer then:

$$\sin(\theta + 2\pi n) = \sin(\theta)$$

$$\csc(\theta + 2\pi n) = \csc(\theta)$$

$$\cos(\theta + 2\pi n) = \cos(\theta)$$

$$\sec(\theta + 2\pi n) = \sec(\theta)$$

$$\tan(\theta + \pi n) = \tan(\theta)$$

$$\cot(\theta + \pi n) = \cot(\theta)$$

Sum and Difference Formulas

$$\sin(\alpha \pm \beta) = \sin(\alpha) \cos(\beta) \pm \cos(\alpha) \sin(\beta)$$

$$\cos(\alpha \pm \beta) = \cos(\alpha) \cos(\beta) \mp \sin(\alpha) \sin(\beta)$$

$$\tan(\alpha \pm \beta) = \frac{\tan(\alpha) \pm \tan(\beta)}{1 \mp \tan(\alpha) \tan(\beta)}$$

Trigonometric Identities

Half Angle Formulas

$$\begin{aligned}\sin\left(\frac{\theta}{2}\right) &= \pm\sqrt{\frac{1 - \cos(\theta)}{2}} \\ \cos\left(\frac{\theta}{2}\right) &= \pm\sqrt{\frac{1 + \cos(\theta)}{2}}\end{aligned} \quad \Rightarrow \quad \tan\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1 - \cos(\theta)}{1 + \cos(\theta)}}$$

Half Angle Formulas (alternate form)

$$\begin{aligned}\sin^2(\theta) &= \frac{1 - \cos(2\theta)}{2} \\ \cos^2(\theta) &= \frac{1 + \cos(2\theta)}{2}\end{aligned} \quad \Rightarrow \quad \tan^2(\theta) = \frac{1 - \cos(2\theta)}{1 + \cos(2\theta)}$$

Calculus on Trigonometric Functions

We know some basic formulas for trigonometric derivatives and antiderivatives:

$$\frac{d}{dx} \sin(x) = \cos(x)$$

$$\frac{d}{dx} \cos(x) = -\sin(x)$$

$$\frac{d}{dx} \tan(x) = \sec^2(x)$$

$$\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$$

$$\frac{d}{dx} \csc(x) = -\csc(x) \cot(x)$$

$$\frac{d}{dx} \cot(x) = -\csc^2(x)$$

$$\int \cos(x) dx = \sin(x) + C$$

$$\int \sin(x) dx = -\cos(x) + C$$

$$\int \sec^2(x) dx = \tan(x) + C$$

$$\int \sec(x) \tan(x) dx = \sec(x) + C$$

$$\int \csc(x) \cot(x) dx = -\csc(x) + C$$

$$\int \csc^2(x) dx = -\cot(x) + C$$

Powers of Sines and Cosines

To integrate $\int \cos^j(x) \sin^k(x) dx$, use the following strategies:

1. If k is odd, rewrite $\sin^k(x)$ as $\sin^{k-1}(x) \sin(x)$ and use the identity $\sin^2(x) = 1 - \cos^2(x)$ to rewrite $\sin^{k-1}(x)$ in terms of $\cos(x)$. Integrate using the substitution $u = \cos(x)$. This substitution makes $du = -\sin(x) dx$.

Powers of Sines and Cosines

To integrate $\int \cos^j(x) \sin^k(x) dx$, use the following strategies:

2. If j is odd, rewrite $\cos^j(x)$ as $\cos^{j-1}(x) \cos(x)$ and use the identity $\cos^2(x) = 1 - \sin^2(x)$ to rewrite $\cos^{j-1}(x)$ in terms of $\sin(x)$. Integrate using the substitution $u = \sin(x)$. This substitution makes $du = \cos(x) dx$. (*Note:* If both j and k are odd, either strategy 1 or strategy 2 may be used.)

Powers of Sines and Cosines

To integrate $\int \cos^j(x) \sin^k(x) dx$, use the following strategies:

3. If both j and k are even, use $\sin^2(x) = \frac{1}{2} - \frac{1}{2} \cos(2x)$ and $\cos^2(x) = \frac{1}{2} + \frac{1}{2} \cos(2x)$. After applying these formulas, simplify and reapply strategies 1 through 3 as appropriate.

Powers of Sines and Cosines

Example: Evaluate $\int \cos^3(x) \sin^2(x) dx$.

Example: Evaluate $\int \sin^2(x) dx$.

Products of $\sin(ax)$ and $\cos(bx)$

$$\sin(ax) \sin(bx) = \frac{1}{2} \cos((a-b)x) - \frac{1}{2} \cos((a+b)x)$$

$$\sin(ax) \cos(bx) = \frac{1}{2} \sin((a-b)x) + \frac{1}{2} \sin((a+b)x)$$

$$\cos(ax) \cos(bx) = \frac{1}{2} \cos((a-b)x) + \frac{1}{2} \cos((a+b)x)$$

Powers of Sines and Cosines

Example: Evaluate $\int \sin^2(x) \cos^2(x) dx$.

Powers of Tangents and Secants

$$\int \sec^2(x) dx = \tan(x) + C$$

$$\int \sec(x) \tan(x) dx = \sec(x) + C$$

$$\int \tan(x) dx = \ln|\sec(x)| + C$$

$$\int \sec(x) dx = \ln|\sec(x) + \tan(x)| + C$$

Integrals involving $\int \tan^j(x) \sec(x)^2 dx$ and $\int \sec^j(x) \tan(x) dx$ can also be done by substitution.

Powers of Tangents and Secants

To integrate $\int \tan^k(x) \sec^j(x) dx$, use the following strategies:

1. If j is even and $j \geq 2$, rewrite $\sec^j(x)$ as $\sec^{j-2}(x) \sec^2(x)$ and use $\sec^2(x) = \tan^2(x) + 1$ to rewrite $\sec^{j-2}(x)$ in terms of $\tan(x)$. Let $u = \tan(x)$ and $du = \sec^2(x) dx$.

Powers of Tangents and Secants

To integrate $\int \tan^k(x) \sec^j(x) dx$, use the following strategies:

2. If k is odd and $j \geq 1$, rewrite $\tan^k(x) \sec^j(x)$ as $\tan^{k-1}(x) \sec^{j-1}(x) \sec(x) \tan(x)$ and use $\tan^2(x) = \sec^2(x) - 1$ to rewrite $\tan^{k-1}(x)$ in terms of $\sec(x)$. Let $u = \sec(x)$ and $du = \sec(x) \tan(x) dx$. (Note: If j is even and k is odd, then either strategy 1 or strategy 2 may be used.)

Powers of Tangents and Secants

To integrate $\int \tan^k(x) \sec^j(x) dx$, use the following strategies:

3. If k is odd where $k \geq 3$ and $j = 0$, rewrite $\tan^k(x)$ as $\tan^{k-2}(x) \sec^2(x) - \tan^{k-2}(x)$. It may be necessary to repeat this process on the $\tan^{k-2}(x)$ term.

$$\begin{aligned}\tan^k(x) &= \tan^{k-2}(x) \tan^2(x) \\ &= \tan^{k-2}(x)(\sec^2(x) - 1) \\ &= \tan^{k-2}(x) \sec^2(x) - \tan^{k-2}(x)\end{aligned}$$

Powers of Tangents and Secants

To integrate $\int \tan^k(x) \sec^j(x) dx$, use the following strategies:

4. If k is even and j is odd, then use $\tan^2(x)$ as $\sec^2(x) - 1$ to express $\tan^k(x)$ in terms of $\sec(x)$. Use integration by parts to integrate odd powers of $\sec(x)$.

Powers of Sines and Cosines

Example: Evaluate $\int \tan^4(x) \sec^4(x) dx$.

Example: Evaluate $\int \tan^3(x) dx$.

Example: Evaluate $\int \sec^3(x) dx$.

Powers of Sines and Cosines

Using integration by parts (as before), we can derive the following “reduction formulas” for powers of secants or tangents.

$$\int \sec^n(x) dx = \frac{1}{n-1} \sec^{n-2}(x) \tan(x) + \frac{n-2}{n-1} \int \sec^{n-2}(x) dx$$

$$\int \tan^n(x) dx = \frac{1}{n-1} \tan^{n-1}(x) - \int \tan^{n-2}(x) dx$$